

CO-4 AT-1

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① Hospital Data Warehouse

(a) Draw a snowflake schema

Fact Table : Consultation - Fact

* Patient - ID

* Doctor - ID

* Dept - ID

* Time - ID

* Count

Dimension Tables

* Patient : Patient - ID, Name, Age, Gender

* Doctor : Doctor - ID, Name, Specialization

* Dept : Dept - ID, Dept - Name

* Time : Time - ID, Day → Month → Quarter

(b) OLAP Operation

To list average consultation fee
department wise for each year:

* slice using measure = Avg - charge

* Group by Dept & Year

(c) number of cuboids

$$(L_1 + 1)(L_2 + 1)(L_3 + 1)(L_4 + 1)$$

Each dimension has 4 levels.

A : 256 cuboids

② Smart city data warehouse

(a) Star schema

Fact Table : Smart-city-Fact
Measures:

- * vehicles-count
- * congestion-level
- * Energy-consumption
- * Ridership

Dimension Tables

- * Day
- * month
- * quarter
- * year.

location

- * Area
- * zone
- * city

Traffic

- * road-type
- * junction.

(b) Measures and Dimensions

Measures

- * vehicle count
- * congestion level
- * Energy consumption
- * Ridership
- * Punctuality

Dimensions

- * Time
- * location
- * transport
- * Traffic
- * Environment.

Banking Data warehouse

(a) Galaxy schema

Fact tables

Transaction - Fact

* Transaction - count

* Amount

Loan - Fact

* Interest

* Loan - Amount

Shared dimensions

* customer

* Branch

* Time

* Account type

(b) OLAP operations

To obtain branch-wise average transaction amount & yearly loan amount

* Roll-up - Time $\rightarrow$ year

* Slice by Branch.

* Aggregate :

(c) Total cuboids

There are 4 dimensions and each has 4 levels

$$4^4 = 256$$

A = 256 cuboids.

④ Manufacturing company

Roll-up

Summarize defect rates by Product category for current year across all factories.

* Roll-up

* Product $\rightarrow$ category

* Time $\rightarrow$ year

Drill-down

Analyze defect rate

* factory $\rightarrow$ Department

* Product $\rightarrow$ Product Type

slice

Select Production unit/
data where

Supplier Rating ≥ 4

Across all regions.

Analyze

* Defect Rate

* Supplier Reliability

⑤ Retail chain

Roll up

Summarize inventory turnover by Product category across all stores for last quarter

slice

Select

* Product category

* factory locations

* Product $\rightarrow$ category

* store $\rightarrow$ All stores

* Time $\rightarrow$ quarter

Drill-down

Analyze inventory in a specific region

Region $\rightarrow$ city $\rightarrow$ store

category $\rightarrow$ subcategory $\rightarrow$ sku

slice

select

membership tier = Bronze

Analyze sales revenue

Across all stores

dice

select

* cities

* Product categories

* time period

Analyze

* sales revenue

* units sold

* sales

formula

Total cuboids

(no of levels in each dimension)

for 4 dimensions with 4 levels each:

$$4 \times 4 \times 4 \times 4 = 256$$

A = 256 cuboids.